A CASE-STUDY BASED REPORT

ON

FEATURE IMPORTANCE ANALYSIS IN DIABETES PREDICTION USING RANDOM FOREST AND XGBOOST

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Abstract

Diabetes mellitus is a chronic disease whose complications can be reduced considerably when it is detected early. This project develops a machine-learning system that classifies patients as diabetic or non-diabetic from routine clinical measurements and, equally important, explains which measurements the models rely on. The study uses the Pima Indians Diabetes dataset, which contains 768 patient records, eight predictor columns and a binary Outcome target; seven predictors were selected for modelling. Two ensemble algorithms, Random Forest and XGBoost, were trained on a stratified 80/20 split and compared using Accuracy, Precision, Recall, F1-score, ROC-AUC and 5-fold cross-validation. On the held-out test set Random Forest obtained 77.3% accuracy and XGBoost 76.0%, while both reached a ROC-AUC of about 0.825; the two models were therefore found to be statistically close, with Random Forest slightly ahead in accuracy and precision and XGBoost ahead in recall and F1-score. Feature-importance analysis ranked Glucose first in both models, followed by BMI and Age. The models were wrapped in an interactive Streamlit application, Diabeta AI, that offers prediction, model analytics, feature-importance visualisation and dataset exploration. The system is an academic demonstration and not a diagnostic tool.

1. Introduction

1.1 Background

Diabetes is one of the most common chronic metabolic diseases in the world. When it remains undiagnosed or poorly controlled it can lead to serious complications such as cardiovascular disease, kidney failure, nerve damage and vision loss. Screening is therefore valuable, but clinical decisions are normally made by looking at several measurements together, such as plasma glucose, body-mass index, blood pressure, insulin level and age, rather than at any single value.

Machine learning offers a way of learning these multi-variable patterns directly from historical patient records. Ensemble methods, which combine many decision trees, are particularly attractive for small tabular medical datasets because they capture non-linear relationships and interactions between variables without heavy feature engineering.

1.2 Motivation for the Project

Most diabetes-prediction projects stop at reporting an accuracy figure. A prediction that cannot be explained is difficult to trust in a healthcare setting, so the motivation of this project is two-fold: **prediction** (can the models separate diabetic from non-diabetic patients?) and **explainability** (which clinical features drive those predictions?).

Random Forest (a bagging ensemble) and XGBoost (a gradient-boosting ensemble) were chosen because both perform strongly on tabular data and both expose built-in feature-importance scores,

which allows the two modelling philosophies to be compared on the same data, the same split and the same metrics.

1.3 Overview of the Work Carried Out

The project follows an end-to-end machine-learning workflow:

Data Collection → Data Preprocessing → Train/Test Split → Model Training → Evaluation → Feature-Importance Analysis → Model Comparison → Web Application

The Pima Indians Diabetes dataset (768 records) was loaded and inspected. Physiologically impossible zero values were treated as missing and imputed with column medians. The data was split in a stratified 80/20 ratio, and a Random Forest (300 trees, balanced class weights) and an XGBoost model (300 estimators, depth 4, learning rate 0.05) were trained. Both were evaluated on the same test partition, supplemented by 5-fold cross-validation, and their built-in feature importances were compared. Finally, the trained models were integrated into a Streamlit web application named **Diabeta AI**.

1.4 Scope and Responsible-Use Statement

The scope of this project is to demonstrate an explainable, ensemble-based diabetes classifier on a standard public benchmark dataset. The system takes seven clinical inputs and returns each model's predicted class and positive-class probability.

The application is an **academic demonstration and is not a diagnostic or clinical decision-support tool**. Predicted probabilities have not been calibrated or externally validated, the dataset is small and describes a specific population, and feature importance describes how a model uses its inputs, not what causes the disease. Users should not enter identifiable or sensitive patient data, and no output of the system should be used for medical decisions.

2. Problem Statement

2.1 Case-Study Scenario

Consider a community health-screening programme that records basic clinical measurements for every visiting adult. Laboratory confirmation for everyone is expensive and slow, so staff would like an inexpensive first-level tool that flags patients with a higher likelihood of diabetes for follow-up testing. Just as importantly, the clinicians who use such a tool need to know why a patient was flagged: a tool that points to high glucose or high BMI is easier to trust than a black box.

The proposed system addresses this scenario by training two ensemble classifiers on historical patient records and by reporting both their predictive performance and the clinical features they depend on most.

2.2 Central Case-Study Question

The central question addressed by this project is:

> ***Can Random Forest and XGBoost use routine clinical measurements to predict whether a patient is diabetic, which of the two performs better, and which clinical features contribute most to their predictions?***

2.3 Formal Problem Definition

Let the input feature vector be:

$$X = \{\text{Pregnancies, Glucose, BloodPressure, Insulin, BMI, DiabetesPedigreeFunction, Age}\}$$

and the target variable be Y = Outcome, where Y = 0 means the patient is non-diabetic and Y = 1 means the patient is diabetic. The objective is to learn a function

$$f(X) \rightarrow P(Y = 1)$$

that returns the estimated probability of diabetes, and to obtain a predicted class by thresholding this probability at 0.5. In addition, for each trained model m the importance vector $I_m(X) = (I_m(x_1), \ldots, I_m(x_7))$ is extracted so that features can be ranked by their contribution to the model's predictions.

2.4 Objectives that Guide the Study

1. Obtain and inspect the Pima Indians Diabetes dataset.
2. Clean the data by handling invalid zero measurements.
3. Split the data into stratified training and testing sets.
4. Train a Random Forest classifier.
5. Train an XGBoost classifier.
6. Compare the models using Accuracy, Precision, Recall, F1-score and ROC-AUC.
7. Validate the comparison using 5-fold cross-validation.
8. Identify the most influential clinical features using feature importance.
9. Interpret the findings without confusing predictive contribution with causation.
10. Build an interactive Streamlit application for prediction, analytics and exploration.

2.5 Assumptions and Limitations

Assumptions

It is assumed that the clinical measurements in the dataset contain predictive information about diabetes status, that the recorded Outcome labels are correct, and that a zero in Glucose, BloodPressure, Insulin or BMI represents a missing measurement rather than a real value.

Limitations

- The dataset is small (768 records) and represents a specific population, so results may not generalise.
- Median imputation was computed on the whole dataset before the train/test split, so the test partition contributed to the imputation values. This is a small preprocessing leakage that should be removed (for example, with a scikit-learn Pipeline) before the scores are treated as rigorous research results.
- The application evaluates the models on a single stratified hold-out split of 154 records; one or two patients change the headline accuracy by roughly 1%.
- Insulin has 374 zero (missing) records, so a large share of its values is imputed.
- Hyper-parameters were fixed and not tuned, and probabilities were not calibrated.
- Feature importance is an association within the trained model and does not prove causation.
- The system performs no input validation as a clinical workflow and is not an official diagnostic tool.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The dataset used in the project is the Pima Indians Diabetes dataset, stored as:

data/diabetes.csv

It contains 768 records and 9 columns: eight predictor columns and one target column. Of the 768 patients, 500 (65.1%) are non-diabetic and 268 (34.9%) are diabetic, so the classes are moderately imbalanced. The dataset has no null cells and no duplicate rows. The columns are described in Table 1.

Feature	Type	Description
Pregnancies	Integer	Number of times the patient has been pregnant
Glucose	Integer	Plasma glucose concentration (oral glucose tolerance test)
BloodPressure	Integer	Diastolic blood pressure (mm Hg)
SkinThickness	Integer	Triceps skin-fold thickness (mm) — not used by the models
Insulin	Integer	Serum insulin (mu U/ml)
BMI	Float	Body-mass index (kg/m²)
DiabetesPedigreeFunction	Float	Score of diabetes family history
Age	Integer	Age in years (21–81)
Outcome (target)	Binary	0 = non-diabetic, 1 = diabetic

Table 1: Description of dataset columns

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) is carried out before modelling to understand the data and to detect problems that would mislead the models. In this project EDA was used to:

- understand the structure, data types and scale of every feature;
- check for null values and duplicate records;
- examine the balance of the target variable;
- detect invalid or physiologically impossible values;
- study how each feature relates to the diabetes outcome and to the other features;
- decide which features and which preprocessing steps are appropriate.

3.3 Preparation Steps Actually Performed

The following steps are implemented in app.py:

Step 1 — Dataset Loading

The CSV file is read into a pandas DataFrame from data/diabetes.csv.

Step 2 — Data Validation

The data was checked for null cells (0) and duplicate rows (0). None were found. However, summary statistics revealed that several columns contain zeros that cannot be real measurements (Figure 2).

Step 3 — Feature Selection

Seven features were selected: Pregnancies, Glucose, BloodPressure, Insulin, BMI, DiabetesPedigreeFunction and Age. SkinThickness is present in the CSV but is not used, because 227 of its 768 values are zero and the remaining features already capture body composition (BMI). Outcome is the target.

Step 4 — Handling Invalid Zeros

Zeros in Glucose, BloodPressure, Insulin and BMI were replaced with NaN and then filled with the median of each column (Glucose 117, BloodPressure 72, Insulin 125, BMI 32.3). The median was chosen over the mean because these columns are right-skewed, particularly Insulin. The same stored medians are applied to any values entered in the prediction form.

Step 5 — Train/Test Split

The data was divided using a stratified split with random_state = 42: 80% training (614 records, 214 diabetic) and 20% testing (154 records, 54 diabetic). Stratification keeps the diabetic/non-diabetic ratio the same in both partitions. No feature scaling was applied because tree-based models are insensitive to monotonic feature scaling.

```
X[missing] = X[missing].replace(0, np.nan)   # 4 columns with invalid zeros
X = X.fillna(X.median())
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.20,
                                      random_state=42, stratify=y)
```

3.4 EDA Exploration — Implementation

The EDA examined the distribution of the target, the presence of invalid values, the separation of the classes by key features, and the correlations among variables.

Target distribution

Figure 1 shows 500 non-diabetic and 268 diabetic patients (a ratio of about 1.9 : 1). The imbalance is moderate; it is the reason Random Forest uses balanced class weights and the reason Precision, Recall and F1-score are reported in addition to accuracy. A model that always predicted "non-diabetic" would already score 65.1% accuracy, so accuracy alone is not sufficient.

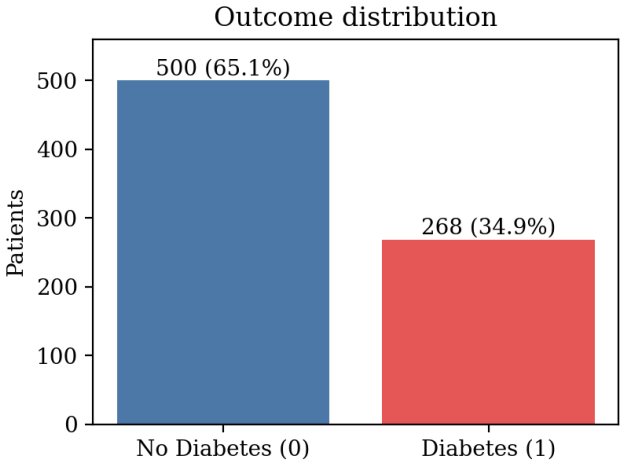


Figure 1: Distribution of the Outcome variable

Invalid zero values

Figure 2 counts the zero values in the columns where zero is physiologically impossible. Insulin (374) and SkinThickness (227) are most affected, followed by BloodPressure (35), BMI (11) and Glucose (5). These zeros are the dataset's way of encoding missing measurements, which is why they were imputed.

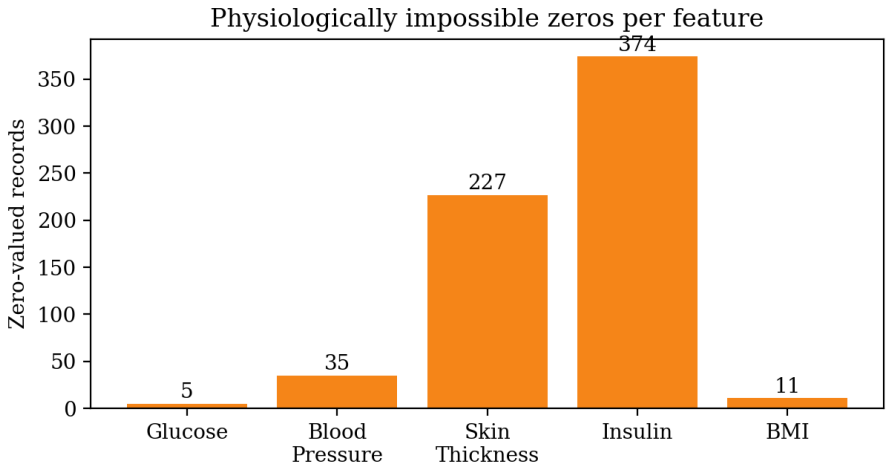


Figure 2: Zero-valued (missing) records per feature

Class separation by glucose

Figure 3 shows that diabetic patients have markedly higher glucose values than non-diabetic patients, with the diabetic distribution shifted to the right. Glucose is the clearest single separator in the data.

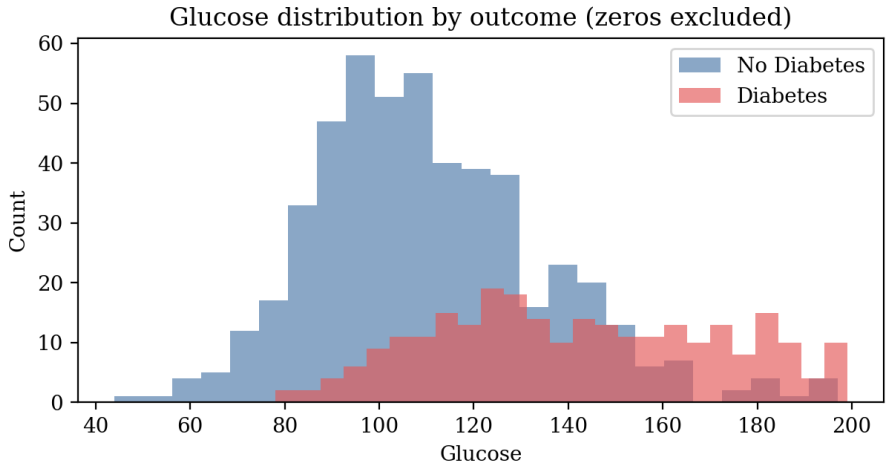


Figure 3: Glucose distribution by outcome

Correlation analysis

Figure 4 and Table 2 show the correlation of each feature with Outcome. Glucose has the strongest linear association (0.467), followed by BMI (0.293), Age (0.238) and Pregnancies (0.222). BloodPressure and SkinThickness have weak correlations. Pregnancies and Age are themselves correlated, which is expected.

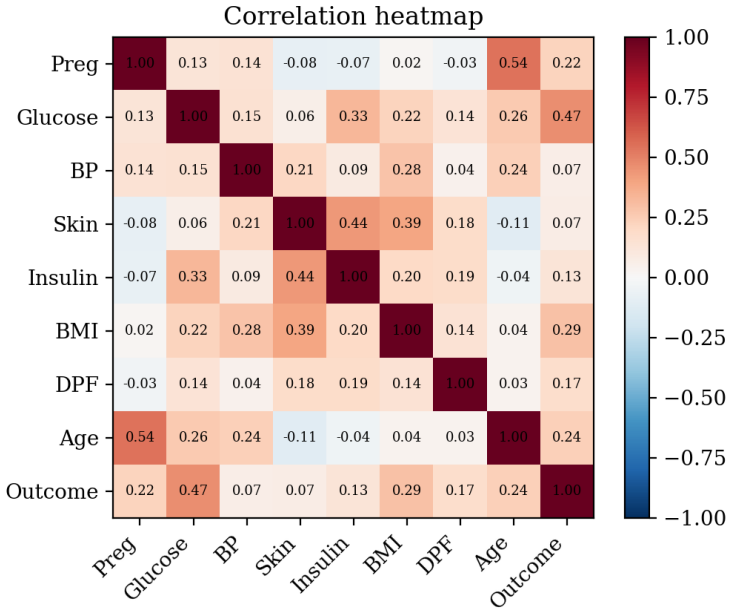


Figure 4: Correlation heatmap of all columns

Feature	Correlation with Outcome
Glucose	0.467
BMI	0.293
Age	0.238
Pregnancies	0.222
DiabetesPedigreeFunction	0.174
Insulin	0.131
SkinThickness	0.075
BloodPressure	0.065

Table 2: Correlation of each feature with Outcome

3.5 Summary of EDA Findings

- The dataset is clean in the technical sense (no nulls or duplicates) but contains many hidden missing values encoded as zeros, particularly in Insulin and SkinThickness.
- The classes are moderately imbalanced (65.1% vs 34.9%), so accuracy must be read together with Precision, Recall, F1-score and ROC-AUC.
- Glucose is the strongest single indicator of diabetes, followed by BMI, Age and Pregnancies.
- The relationship between features and outcome is not purely linear and involves interactions, which supports the use of tree-based ensemble models.
- No feature scaling is required for the chosen algorithms, but median imputation is necessary.

4. Algorithm Selection and Implementation

Two ensemble-learning algorithms were selected so that a bagging approach and a boosting approach could be compared on exactly the same data, split and metrics.

4.1 Random Forest

Random Forest builds many decision trees, each trained on a different bootstrap sample of the training data and using a random subset of features at each split. The final class is decided by the combined vote of all the trees. Because the trees are built independently and averaged, the method reduces variance, is robust to noise and outliers, handles non-linear relationships and needs relatively little tuning. It also provides a built-in (Gini) feature-importance score. The project used:

```
RandomForestClassifier(n_estimators=300, random_state=42,
                       class_weight="balanced")
```

class_weight="balanced" increases the weight of the minority (diabetic) class to compensate for the 65/35 class imbalance.

4.2 XGBoost

XGBoost is an optimised gradient-boosting method. Trees are built sequentially, and each new tree is trained to correct the errors made by the trees before it. A small learning rate, shallow trees and row/column subsampling control over-fitting. XGBoost generally performs very strongly on tabular data and provides its own feature-importance scores. The project used:

```
XGBClassifier(n_estimators=300, max_depth=4, learning_rate=0.05,
              subsample=0.85, colsample_bytree=0.85,
              eval_metric="logloss", random_state=42)
```

4.3 Model Evaluation

Both models were fitted on the 614 training records and evaluated on the same 154 test records. Accuracy, Precision, Recall, F1-score and ROC-AUC were computed with scikit-learn. As a supplementary robustness check, stratified 5-fold cross-validation (shuffled, random_state = 42) was also run on the imputed dataset using the same model settings. Results are shown in Table 3.

Metric	Random Forest	XGBoost
Accuracy	77.3%	76.0%
Precision	72.1%	66.7%
Recall	57.4%	63.0%
F1-score	63.9%	64.8%
ROC-AUC (test)	0.825	0.826
5-fold CV ROC-AUC (mean ± SD)	0.831 ± 0.024	0.821 ± 0.031
5-fold CV Accuracy (mean ± SD)	77.3% ± 2.3	75.0% ± 2.5

Table 3: Test-set performance and 5-fold cross-validation results

The confusion matrices on the 154 test patients are: Random Forest — 88 true negatives, 12 false positives, 23 false negatives, 31 true positives; XGBoost — 83 true negatives, 17 false positives, 20 false negatives, 34 true positives.

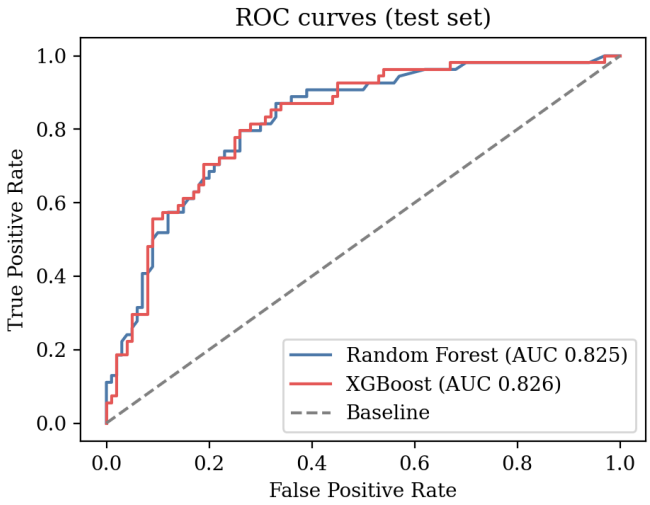


Figure 5: ROC curves of both models on the test set

Interpretation

- **Neither model dominates.** Random Forest has higher accuracy (77.3% vs 76.0%), higher precision (72.1% vs 66.7%) and a better cross-validated AUC (0.831 vs 0.821). XGBoost has higher recall (63.0% vs 57.4%) and F1-score (64.8% vs 63.9%).
- The test ROC-AUC values are practically identical (0.825 vs 0.826), so both rank patients equally well. Differences of one to three patients on a 154-record test set are within normal sampling noise, and the cross-validation standard deviations (about 2–3 points) are larger than the gaps between the models.
- XGBoost detected 34 of the 54 diabetic test patients against 31 for Random Forest, but at the price of five more false alarms. For a screening tool where missing a diabetic patient is costlier than an unnecessary follow-up test, XGBoost's higher recall is attractive; if fewer false alarms matter more, Random Forest is preferable.
- Both models achieve only moderate recall (57–63%), so a substantial fraction of diabetic patients is still missed. Threshold tuning and probability calibration would be the next steps.

4.4 Feature-Importance Results

Built-in feature importances were extracted from both trained models (Gini importance for Random Forest and XGBoost's default importance for the boosted trees). The values are listed in Table 4 and plotted in Figure 6.

Feature	Random Forest	XGBoost	RF Rank	XGB Rank
Glucose	0.2750	0.2688	1	1
BMI	0.1830	0.1513	2	2
Age	0.1402	0.1386	3	3
DiabetesPedigreeFunction	0.1301	0.1092	4	5
Insulin	0.1063	0.1340	5	4
BloodPressure	0.0866	0.0936	6	7
Pregnancies	0.0789	0.1045	7	6

Table 4: Feature importance from Random Forest and XGBoost

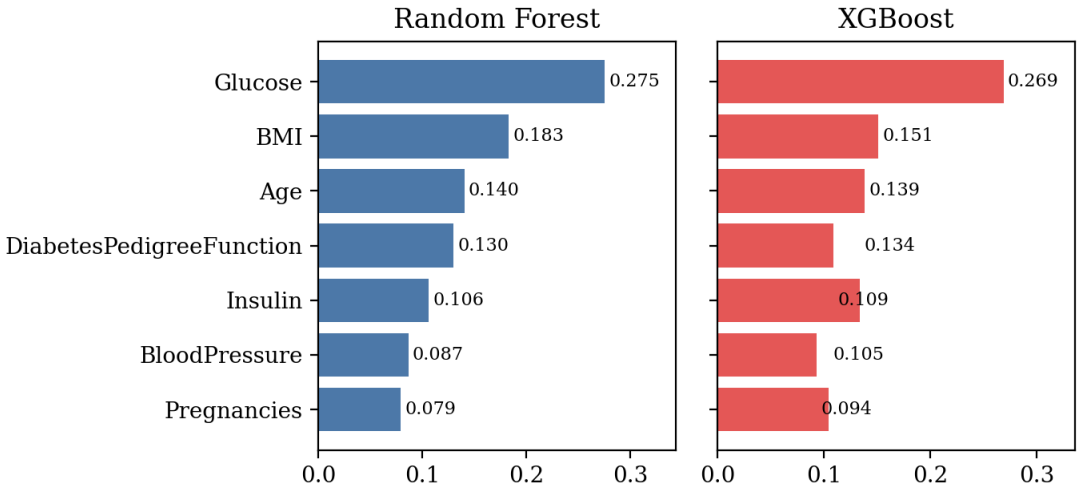


Figure 6: Feature importance of Random Forest and XGBoost

Glucose is the most important feature in both models (0.275 and 0.269), contributing roughly one quarter of the total importance, which agrees with the EDA. BMI and Age rank second and third in Random Forest; in XGBoost BMI is second and Age third, with Insulin close behind. The two models therefore agree on the top three features, which gives confidence in the finding. Importance is spread fairly evenly over the remaining features, so no feature can be dropped without some loss of information.

Important note: a high importance score means that the feature contributes strongly to the model's predictions; it does not mean that the feature causes diabetes. In addition, impurity-based importance can favour features with many distinct values and is less reliable when features are correlated.

4.5 Implementation and Deployment

The trained models are deployed in a Streamlit web application called Diabeta AI (app.py). The application trains both models when it starts, so every metric, chart and prediction shown is computed from the fitted models and none is hard-coded. The interface has six pages:

Page	Function
Dashboard	Record count, number of features, ROC-AUC of both models, outcome distribution and metric comparison charts, project workflow
Prediction	Form for the seven clinical inputs; shows each model's predicted class (higher/lower predicted risk) and probability of Outcome = 1
Model Analytics	Metrics table, confusion matrices and ROC curves for both models
Feature Importance	Side-by-side importance bar charts and a ranked table
Dataset Explorer	Data preview, missing-cell count, histograms, scatter plot and descriptive statistics
About	Objective, models and technology summary with a disclaimer

Table 5: Pages of the Diabeta AI application

Software requirements. Python 3.10 or later with Streamlit, pandas, NumPy, scikit-learn, XGBoost and Plotly (listed in requirements.txt). The application is started with the command: python -m streamlit run app.py and opens in the browser at http://localhost:8501.

When a user submits the prediction form, any zero entered for Glucose, BloodPressure, Insulin or BMI is replaced by the stored training median, the input is passed to both models, and the predicted class and probability are displayed side by side together with a prominent "educational/research project only" warning.

5. Conclusion

This project developed an end-to-end, explainable machine-learning system for diabetes prediction. Using the Pima Indians Diabetes dataset of 768 patients and seven clinical features, a Random Forest and an XGBoost classifier were trained, evaluated and compared using Accuracy, Precision, Recall, F1-score, ROC-AUC and 5-fold cross-validation.

The two ensembles performed almost identically. Random Forest reached 77.3% accuracy with a precision of 72.1% and the better cross-validated AUC (0.831), while XGBoost reached 76.0% accuracy with the better recall (63.0%) and F1-score (64.8%); both achieved a test ROC-AUC of about 0.825. The choice between them therefore depends on whether false alarms or missed cases are considered more costly, rather than on a clear overall winner.

The feature-importance analysis was consistent across both models: **Glucose is the most influential feature, followed by BMI and Age**, matching the correlation analysis from the EDA. These scores describe predictive contribution within the models and must not be read as proof of causation. By combining prediction with interpretability, and by delivering both through the Diabeta AI application, the project shows how machine learning can support, but not replace, diabetes screening.

The current implementation is an academic prototype. Its limitations — a small single-population dataset, median imputation done before the split, a single hold-out evaluation, untuned hyper-

parameters, uncalibrated probabilities and no external validation — mean that it must not be used for diagnosis or any medical decision.

Future work can include leakage-safe scikit-learn pipelines, hyper-parameter tuning, repeated cross-validation, probability calibration and threshold analysis, local explanation methods such as SHAP, validation on additional datasets, input validation, and automated testing.

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